

MODULE 3A · EPISODE 2

Making Measurements

Significant Figures, Precision, and Units

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Significant Figures, Precision, and Units

This episode picks up where the last one left off and asks how to write a measurement honestly. The panel starts with significant figures — the digits you know for sure plus one guessed digit — and shows why scientific notation clears up the confusion caused by leading and trailing zeros. The discussion then turns to the difference between precision and accuracy, and to systematic errors: trivial ones like zero error and parallax, which should simply be corrected, and non-trivial ones built into the equipment or procedure. The second half covers units: the US-English and metric systems, the MKS and CGS varieties of the metric system, and the conversion factors that link them. The episode closes with the rules for carrying significant figures through calculations, so that a calculator's ten digits never claim more precision than the measurements justify.

The Panel

Name	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Guest; cosmologist, European Space Agency
Dr. Amara Johnson	Guest; theoretical physicist

Learning Objectives

After this episode you should be able to:

- Define significant figures and identify the least significant digit of a measurement.
- Write a measurement in scientific notation and explain why it removes ambiguity.
- Count significant figures, including leading and trailing zeros.
- Distinguish precision from accuracy, and explain why precision is necessary but not sufficient for accuracy.
- Tell trivial systematic errors (zero error, parallax) from non-trivial ones, with an example of each.
- Explain parallax and use proper technique to minimize it when reading analog devices.
- Name the US-English and MKS units for length, time, mass, energy, force, and charge.
- Use conversion factors within and between unit systems.
- Round the result of a calculation to the correct number of significant figures.

QUICK REFERENCE

Key Terms

Term	Definition
Significant figures (sig figs)	All the digits in a measurement that you know for sure, plus one more digit that is guessed or uncertain.
Least significant digit	The rightmost significant digit; the one that was guessed, or uncertain.
Scientific notation	A number between 1 and 10, not including 10, multiplied by the necessary power of ten.
Standard notation	A number written out in ordinary form, such as 0.0069 or 75,000,000.
Leading zeros	Zeros to the left of the first nonzero digit. In standard notation they are never significant; they are placeholders.
Trailing zeros	Zeros at the right end of a number. In standard notation they may or may not be significant; in scientific notation they always are.
Accuracy	How close a measurement is to the true value.
Precision	How finely you can measure (how many significant figures you can legitimately write) and how closely repeated measurements agree.
Systematic error	An error that pushes measurements consistently in one direction.
Trivial systematic error	Only nominally a systematic error; really a mistake or blunder that should be corrected, such as zero error or parallax.
Zero error	An error introduced when a device does not read zero when it should.
Non-trivial systematic error	An inherent limitation in the equipment or procedure of an experiment; it can sometimes be accounted for, but not really corrected.
Reaction-time and anticipation errors	Errors built into hand timing: starting or stopping a timer a little late, or a little early if you anticipate the signal.
Parallax	The apparent shift of a nearby object against a more distant background when the viewing position, or the eye used, changes.
Physical unit	A standardized amount of a physical quantity.
US-English system	The system of units (feet, pounds, seconds) used almost exclusively in the United States; sometimes mistakenly called the Imperial system.
MKS system	The metric system based on meters, kilograms, and seconds; the basis of the SI. Used in physics.
CGS system	The metric system based on centimeters, grams, and seconds. Common in chemistry.
SI	The International System of Units, from the French <i>Système International</i> .
Slug / pound-mass	Two competing units of mass used in the US-English system.
Conversion factor	A ratio equal to the number one, used to change one unit to another.
Linking conversion factor	A conversion factor that connects the US-English and metric systems, such as 2.54 cm per 1 inch.
Linear mass density	Mass per unit length, $D = m/L$.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
4	Significant digits in 12.34 cm; the 4 is the least significant digit
2	Significant digits in 0.0069 cm (the 6 and the 9)
75,000,000 yr	Dinosaur-bone age in standard notation; the trailing zeros are ambiguous
7.50×10^7 yr	The same age in scientific notation: three significant figures, final zero uncertain
12.34 ± 0.01 cm	First measurement in the precision example
12.345 ± 0.001 cm	More precise measurement, but not necessarily more accurate
12 in = 1 ft	Also 3 ft = 1 yd and 5,280 ft = 1 mi
1000	mm per m, m per km, and g per kg in the metric system
2.54 cm = 1 in	Linking conversion factor
39.37 in = 1 m	Linking conversion factor
1.609 km = 1 mi	Linking conversion factor
2.205 lb = 1 kg	Convenient, but not “kosher”: it turns a unit of force into a unit of mass
60 / 60 / 24	Seconds per minute, minutes per hour, hours per day
365.25 days = 1 yr	The Julian calendar approximation
12.3 g, 2.964 cm	Mass (3 sig figs) and length (4 sig figs) of the thin rod
4.149797571 g/cm	Calculator result for $D = m/L$: ten digits, most of them meaningless
4.15 g/cm	The correctly reported linear mass density

EPISODE CONTENT

Part-by-Part Summary

Part 1 — Significant Figures

Significant figures, also called significant digits or “sig figs,” are all the digits you know for sure, plus one more digit that you guess or that is in some way uncertain. If your best guess for a measurement is 12.34 cm, there are four significant digits. The rightmost digit, the 4, is the **least significant digit**: the one that was guessed.

Significant figures tell us about the **precision** of a measurement — how many meaningful digits we can legitimately claim to know. Using them properly takes practice and patience.

The first and most important rule: express measurements in **scientific notation**, as a number between 1 and 10 (not including 10) multiplied by the necessary power of ten. **Standard notation** is often confusing. **Leading zeros** in standard notation are never significant: 0.0069 cm has only two significant digits, the 6 and the 9. The zeros are just placeholders.

Part 2 — Trailing Zeros and Scientific Notation

Trailing zeros in standard notation may or may not be significant. If a dinosaur bone is reported to be 75,000,000 years old, how many of those zeros were actually measured, and how many are just placeholders? There is no way to tell.

Trailing zeros in scientific notation are always significant, because otherwise you simply wouldn't write them. Writing the bone's age as 7.50×10^7 years states explicitly that there are three significant figures and that the zero after the 5 is the guessed, least significant digit. Scientific notation removes the ambiguity.

Part 3 — Precision and Accuracy

Accuracy is how close a measurement is to the true value. **Precision** is how finely you can measure — how many significant figures you can legitimately write — and how closely repeated measurements agree with one another. The two are often confused.

It is tempting to think that 12.345 ± 0.001 cm must be closer to the true value than 12.34 ± 0.01 cm because its confidence interval is smaller. Not necessarily: a very precise measurement can be quite inaccurate if the device was not properly calibrated — for example, if it has an uncorrected **zero error**.

The correct relationship: being precise is **necessary, but not sufficient**, for being accurate. A device that is systematically off by a fixed amount gives precise measurements that are all consistently wrong — high precision, low accuracy. A calibrated but imprecise device gives measurements that scatter around the true value — better accuracy on average, but lower precision.

Part 4 — Systematic Errors and Parallax

There are two main categories of **systematic errors**. A **trivial systematic error**, such as a zero error, is only nominally a systematic error; it is really a mistake, a blunder, and should be corrected as a matter of course.

A **non-trivial systematic error** is an inherent limitation in the equipment or procedure. It can't really be corrected, though it can sometimes be accounted for in the analysis. Starting and stopping a timer by hand builds **reaction-time and anticipation errors** into every time measurement: you may consistently be a little late responding to a voice command, or a little early if you anticipate it.

Parallax, or eyeball error, is another trivial systematic error. Nearby objects appear to shift against more distant ones depending on where you view them from, including which eye you use: hold your thumb at arm's length, close each eye in turn, and your thumb jumps against the wall. Parallax only affects **analog** devices read by eye, where an edge or needle is compared with a ruled scale.

To minimize it, line your nose up directly over the measurement and alternate eyes; the reading should shift the same amount to each side. With “one good eye,” use that eye directly over the measurement, looking straight down at the scale. Above all, develop a consistent technique.

Part 5 — Units and Systems of Units

A **physical unit** is a standardized amount of a physical quantity. Units let people agree on how much of something was bought at a grocery store, and the same principle applies to experiments: without standardized units, scientific communication would be impossible.

Two main systems are in use today. The **US-English system**, sometimes mistakenly called the Imperial system, is used almost exclusively in the United States, with some lingering use of the Imperial system in the United Kingdom. The **metric system** comes in two varieties: **MKS** (meters, kilograms, seconds), the basis of the **SI**, used almost exclusively in physics; and **CGS** (centimeters, grams, seconds), whose grams and cubic centimeters are common in chemistry.

The most frequently measured quantities are length (feet vs. meters), time (seconds in both), mass (kilograms in MKS), energy (foot-pounds vs. joules), force (pounds vs. newtons), and charge (coulombs in both). The US-English system was designed around force rather than mass, so the pound is technically a unit of force. Some textbooks use the **slug** for mass, others the **pound-mass** — one more reason the metric system is preferred in science.

Part 6 — Conversion Factors and Calculations

A **conversion factor** is a ratio equal to the number one that changes one unit to another: 12 inches per 1 foot, 3 feet per 1 yard, 5,280 feet per 1 mile; 1000 millimeters per 1 meter, 1000 meters per 1 kilometer, 1000 grams per 1 kilogram. The metric system is simpler because it is all based on powers of 10.

Linking conversion factors connect the two systems: 2.54 cm per 1 inch, 39.37 inches per 1 meter, 1.609 km per 1 mile. The factor 2.205 pounds per 1 kilogram is not considered “kosher,” because it changes a unit of force into a unit of mass, but it is very convenient, so we use it. Time factors are shared by both systems: 60 s per minute, 60 min per hour, 24 h per day, and 365.25 days per year, the Julian calendar approximation.

A thin rod has a mass of 12.3 g (three significant figures) and a length of 2.964 cm (four). Its **linear mass density** $D = m/L$ comes out on a calculator as 4.149797571 g/cm. But a calculator can't create

significant digits out of thin air: the uncertainty in each measurement's least significant digit “taints” all but the leftmost digits of the result, so the answer is $D = 4.15 \text{ g/cm}$.

For **multiplication and division**, the result can have no more significant figures than the measurement with the fewest. For **addition and subtraction**, the result has the same number of decimal places as the measurement with the fewest decimal places. These rules keep us honest about the limitations of our data.

The next episode ventures into astronomical measurements, where the scales and techniques are quite different from measurements here on Earth.

TRY IT YOURSELF

Measure the length and width of a sheet of notebook paper with a ruler marked in millimeters, and write each as a scientific measurement, such as $279.5 \pm 0.5 \text{ mm}$. Multiply them to find the area, and round the calculator's answer to the correct number of significant figures. Then convert the length to inches using 2.54 cm per 1 inch.

STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Significant Figures

- 1.** What are significant figures (also called significant digits or “sig figs”)?

- 2.** A measurement is recorded as 12.34 centimeters. How many significant digits does it have, and which digit is the least significant digit?

- 3.** What do the significant figures of a measurement tell us?

- 4.** What is scientific notation, and why should measurements be expressed in it?

5. How many significant digits are in 0.0069 centimeters? Explain.

Part 2 — Trailing Zeros and Scientific Notation

1. Why are trailing zeros in standard notation confusing? Use the example of a dinosaur bone reported to be 75,000,000 years old.

2. Are trailing zeros in scientific notation significant? Why?

3. If the bone's age is written as 7.50×10^7 years, how many significant figures does it have, and which digit is the uncertain one?

Part 3 — Precision and Accuracy

1. Define accuracy.

2. Define precision. What two ideas does it include?

3. A length is first recorded as 12.34 ± 0.01 cm, and later, with a more precise device, as 12.345 ± 0.001 cm. Must the second measurement be closer to the true value? Explain.

4. What is the correct way to describe the relationship between precision and accuracy?

5. Describe a situation with high precision but low accuracy, and one with better accuracy but lower precision.

Part 4 — Systematic Errors and Parallax

1. What are the two main categories of systematic errors?

2. What is a trivial systematic error? Give an example and explain what should be done about it.

3. What is a non-trivial systematic error? Describe the example given in the episode.

4. What is parallax? Describe the thumb demonstration.

5. When does parallax error occur in measurement, and what factors affect it?

6. How can parallax errors be minimized?

Part 5 — Units and Systems of Units

1. What is a physical unit, and why are units necessary?

2. What are the two main systems of units in use today, and where is each used?

3. What are the two varieties of the metric system, and which sciences tend to use each?

4. Give the US-English and MKS units for length, time, mass, energy, force, and charge.

5. Why was no unit of mass given for the US-English system?

Part 6 — Conversion Factors and Calculations

1. What is a conversion factor? Give examples within the US-English system and within the MKS metric system.

2. Name three linking conversion factors between the two systems. Why is 2.205 pounds per 1 kilogram not considered “kosher”?

3. List the time conversion factors given in the episode. What does 365.25 days per 1 year represent?

4. A thin rod has a mass of 12.3 grams and a length of 2.964 centimeters. A calculator gives $D = m/L = 4.149797571$ grams per centimeter. What is the correct answer, and why?

5. State the significant-figure rules for (a) multiplication and division and (b) addition and subtraction.

6. Why do these significant-figure rules matter?

Notes

[illegible]